

Rayleigh wave transport along a surface with random impedance

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ABSTRACT: In this paper we derive a radiative transfer equation for Rayleigh surface waves on the flat surface of an elastic half-space with a random boundary impedance. This transport equation applies to the surface wave energy density resolved in phase space. The analysis uses some classical pseudo-differential calculus applied to a steady-state formulation. Explicit expressions of the scattering cross-section for Rayleigh waves and their leakage rate to bulk waves within the half-space are obtained as functions of the power spectrum of the random impedance. The influence of the latter on the scattering and radiation mean free paths is studied numerically. This problem has interest for applications to non-destructive monitoring and evaluation of structural components, seismic site effects in geophysics, passive imaging, or civil engineering among several other significant examples.

1 INTRODUCTION

The propagation of wave energy in random media can be analyzed with a radiative transfer equation in the case of weak random fluctuations and propagation distances/times that are long with respect to the small wavelength/period of the signal (high-frequency limit). A systematic and efficient way to derive the transport regime from random wave equations is presented by Papanicolaou & Ryzhik (1999), among others. Here we focus on the case of elastic (Rayleigh) surface waves on a randomly inhomogeneous surface in this very limit. This issue is a typical example of high-frequency wave propagation in a two-dimensional randomly heterogeneous medium. A classical result in quantum physics is to consider that all states in two-dimensional disordered systems are localized (Imry 1997). However it is observed for classical waves that the localization length can be much larger than the scattering mean free path, such that transport dominates over localization in a broad range of distances. The basic objective of this research is to derive such transport properties, considering more particularly elastic surface waves. Multiple scattering of the Rayleigh waves propagating on the flat surface of an elastic half-space with a random boundary impedance may be described by a transport model as outlined above. In this respect, the methodology of Bal et al. (2000) or Bal and Ryzhik (2002) can be extended to the case of vector surface waves in order to derive a radiative transfer equation for their amplitude. The latter applies to the associated energy density resolved in phase space. The analysis uses some classical pseudo-differential calculus and considers a steady-state formulation. It also yields explicit expressions of the scattering cross-section for those surface waves and their leakage rate to bulk waves within the half-space. These parameters classically describe the so-called mesoscopic regime of wave

propagation phenomena; they have a direct physical interpretation in terms of scattering and radiation mean free paths. For surface waves they can be obtained as functions of the power spectrum of the random impedance. A formal computation of the scattering parameters for Rayleigh waves has however not been reported so far, to the author's knowledge. The main purpose of this paper is thus to propose a possible theoretical derivation of such properties. Their study is of broad interest for applications to seismic site effects in earthquake engineering (Lombaert and Clouteau 2006), SAW imaging techniques (White & Voltmer 1965), passive imaging techniques (Shapiro et al. 2005), or civil engineering and non-destructive evaluation of structural components at large, among several other significant examples.

In the following section, we first introduce the basic equations describing the propagation of high-frequency Rayleigh waves and the notations used throughout the paper. The short Sect. 3 introduces the main mathematical tools used in the subsequent analysis of the contemplated problem, which is formalized as a two-scale boundary condition for the amplitude vector of the Rayleigh waves. The main results of this research are outlined in the Sect. 4, where a detailed derivation of the corresponding radiative transport equation is provided together with a short discussion of the influence of the random impedance power spectrum on the scattering parameters (scattering and radiation mean free paths).

2 HIGH-FREQUENCY RAYLEIGH WAVES SETTING

2.1 Notations and hypotheses

A Rayleigh wave $\mathbf{u} = (u_x, u_y, u_z)$ propagating at the frequency $\omega > 0$ in an homogeneous, isotropic elastic

half-space $(\mathbf{x}, z) \in \mathbb{R}^2 \times \mathbb{R}_+$ of which surface Γ is $\{z = 0\}$ reads as:

$$\begin{pmatrix} u_x \\ u_y \\ u_z \end{pmatrix} = e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} \mathbf{Q}(\mathbf{k}, z) \begin{pmatrix} A_p \\ A_{s1} \\ A_{s2} \end{pmatrix}, \quad (1)$$

where:

$$\mathbf{Q}(\mathbf{k}, z) = \begin{bmatrix} \hat{k}_x e^{i\phi_p z} & \hat{k}_x e^{i\phi_s z} & -i\hat{k}_y e^{i\phi_s z} \\ i\hat{k}_y e^{i\phi_p z} & i\hat{k}_y e^{i\phi_s z} & \hat{k}_x e^{i\phi_s z} \\ \frac{i\phi_p}{k} e^{i\phi_p z} & \frac{k}{i\phi_s} e^{i\phi_s z} & 0 \end{bmatrix},$$

$i = \sqrt{-1}$, and $\mathbf{k} = k\hat{\mathbf{k}}$ with $k = \sqrt{k_x^2 + k_y^2}$ is the horizontal wave vector such that $\hat{\mathbf{k}} \cdot \hat{\mathbf{n}} = 0$ if $\hat{\mathbf{n}}$ is the outward unit normal to the half-space. Note that the unit wave vector $\hat{\mathbf{k}}$ lies on the unit circle S^1 of $\mathbb{R}^2$. The vertical wavenumbers $\phi_\alpha(k)$ for either compressional ($\alpha = p$) or transverse ($\alpha = s$) motions are defined by:

$$\phi_\alpha(k) = \begin{cases} \sqrt{k_\alpha^2 - k^2} & \text{if } k \leq k_\alpha, \\ i\sqrt{k^2 - k_\alpha^2} & \text{if } k > k_\alpha. \end{cases}$$

Here $k_\alpha = \frac{\omega}{c_\alpha}$ and wave celerities c_α are given as functions of the half-space material density ϱ and Lamé's moduli λ, μ by:

$$c_p = \sqrt{\frac{\lambda + 2\mu}{\varrho}}, \quad c_s = \sqrt{\frac{\mu}{\varrho}}.$$

Coefficients A_p, A_{s1} and A_{s2} are derived from the traction-free boundary condition on Γ :

$$\mathbf{t}(\mathbf{u}) := \lambda(\nabla \cdot \mathbf{u})\hat{\mathbf{n}} + 2\mu(\nabla \otimes_s \mathbf{u})\hat{\mathbf{n}}|_{z=0} = \mathbf{0},$$

which also reads as:

$$\mu e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} \mathbf{P}(\mathbf{k}) \begin{pmatrix} A_p \\ A_{s1} \\ A_{s2} \end{pmatrix} = \mathbf{0}, \quad \forall \mathbf{x} \in \Gamma, \forall t \in \mathbb{R}, \quad (2)$$

where:

$$\mathbf{P}(\mathbf{k}) = \begin{bmatrix} 2k_x \frac{\phi_p}{k} & k_x \left(\frac{\phi_s}{k} - \frac{k}{\phi_s} \right) & -k_y \frac{\phi_s}{k} \\ 2k_y \frac{\phi_p}{k} & k_y \left(\frac{\phi_s}{k} - \frac{k}{\phi_s} \right) & k_x \frac{\phi_s}{k} \\ \frac{k_s^2 - 2k^2}{k} & -2k & 0 \end{bmatrix}.$$

The solvability condition for the above linear system yields Rayleigh's secular equation:

$$(2k^2 - k_s^2)^2 + 4k^2 \phi_p(k) \phi_s(k) = 0, \quad (3)$$

which has only one positive solution k_R^2 such that $k_R > k_s > k_p$. Introducing notations $\phi_p(k) = ip$ and $\phi_s(k) = iq$ for $k > k_s$, the above equation is also:

$$(q^2 + k^2)^2 = 4k^2 pq. \quad (4)$$

The associated amplitudes are derived as:

$$A_{s1}(\mathbf{k}) = -\frac{2pq}{q^2 + k^2} A_p(\mathbf{k}), \quad A_{s2}(\mathbf{k}) = 0.$$

2.2 High-frequency boundary condition with random impedance

Instead of the usual traction-free condition $\mathbf{t}(\mathbf{u}) = \mathbf{0}$ on Γ , we shall consider a more general boundary condition of the form (see e.g. Bal et al. 2000, Garova et al. 1999):

$$(\mathbf{t}(\mathbf{u}), \hat{\mathbf{n}}) = \mu Z(\mathbf{x})(\mathbf{u}, \hat{\mathbf{n}}) \text{ on } \Gamma, \quad (5)$$

where $(Z(\mathbf{x}), \mathbf{x} \in \mathbb{R}^2)$ is a second-order, mean-square homogeneous and centered stochastic process with real values and correlation function R :

$$R(\mathbf{x} - \mathbf{y}) = \mathbb{E}\{Z(\mathbf{x})Z(\mathbf{y})\},$$

and $\mathbb{E}$ stands for the ensemble average (mathematical expectation). It is assumed that its fluctuations are weak, that is:

$$\varepsilon := \frac{R(\mathbf{0})}{k_R^2} \ll 1, \quad (6)$$

and that the wavelength and correlation length are of the same order ε . Then the boundary condition reads in the rescaled variables $\mathbf{x} \rightarrow \frac{\mathbf{x}}{\varepsilon}, z \rightarrow \frac{z}{\varepsilon}$ as:

$$\varepsilon(\mathbf{t}(\mathbf{u}_\varepsilon), \hat{\mathbf{n}}) = \sqrt{\varepsilon} \mu Z\left(\frac{\mathbf{x}}{\varepsilon}\right)(\mathbf{u}_\varepsilon, \hat{\mathbf{n}}) \text{ on } \Gamma. \quad (7)$$

The amplitude $\sqrt{\varepsilon}$ of the random impedance is consistent with Eq. (6) above. We also assume that steady-state solutions $\hat{\mathbf{u}}_\varepsilon$ of Eq. (7) have the form:

$$\hat{\mathbf{u}}_\varepsilon(\mathbf{x}, z) = \int_{\mathbb{R}^2} e^{i\frac{\mathbf{k} \cdot \mathbf{x}}{\varepsilon}} \mathbf{Q}\left(\mathbf{k}, \frac{z}{\varepsilon}\right) \mathbf{b}(\mathbf{k}) \hat{A}_\varepsilon(\mathbf{k}) \frac{d\mathbf{k}}{(2\pi)^2}$$

where:

$$\mathbf{b}(\mathbf{k}) = \begin{pmatrix} -i \\ \frac{2ipq}{q^2 + k^2} \\ 0 \end{pmatrix},$$

$\hat{A}_\varepsilon \in \mathbb{C}$, and matrix $\mathbf{Q}$ is given by Eq. (1). Introducing the functions $\mathcal{Q}(\mathbf{k}) = (\mathbf{Q}(\mathbf{k}, 0)\mathbf{b}(\mathbf{k}), \hat{\mathbf{n}})$ and $P(\mathbf{k}) = (\mathbf{P}(\mathbf{k})\mathbf{b}(\mathbf{k}), \hat{\mathbf{n}})$ given for $k = |\mathbf{k}|$ by:

$$\mathcal{Q}(\mathbf{k}) = \frac{k_s^2 \phi_p(k)}{k(2k^2 - k_s^2)} \quad (8)$$

and:

$$P(\mathbf{k}) = i\Omega(\mathbf{k}) = i \left[\frac{(2k^2 - k_s^2)^2 + 4k^2 \phi_p(k) \phi_s(k)}{k(k_s^2 - 2k^2)} \right], \quad (9)$$

respectively, with $\mathbf{P}(\mathbf{k})$ defined by Eq. (2), the boundary condition of Eq. (7) finally reads:

$$P(\varepsilon \mathbf{D}) A_\varepsilon(\mathbf{x}) = \sqrt{\varepsilon} Z\left(\frac{\mathbf{x}}{\varepsilon}\right) \mathcal{Q}(\varepsilon \mathbf{D}) A_\varepsilon(\mathbf{x}). \quad (10)$$

In the above $P(D)$ and $Q(D)$ are the pseudo-differential operators on $\mathcal{S}(\mathbb{R}^2)$ (the Schwartz space of all C^∞ functions on $\mathbb{R}^2$ which are rapidly decreasing toward 0 at infinity as well as all their derivatives) with symbols $P(\mathbf{k})$ and $Q(\mathbf{k})$, respectively. Here the standard form of the Fourier transform in $L^2(\mathbb{R}^2)$ is:

$$\widehat{f}(\mathbf{k}) = \int_{\mathbb{R}^2} e^{-i\mathbf{k}\cdot\mathbf{x}} f(\mathbf{x}) d\mathbf{x},$$

$$f(\mathbf{x}) = \int_{\mathbb{R}^2} e^{i\mathbf{k}\cdot\mathbf{x}} \widehat{f}(\mathbf{k}) \frac{d\mathbf{k}}{(2\pi)^2},$$

and the pseudo-differential operator $K(\mathbf{x}, D)$ of symbol $K(\mathbf{x}, \mathbf{k})$ (a smooth complex-valued function of both arguments) is:

$$K(\mathbf{x}, \varepsilon D)f(\mathbf{x}) = \int_{\mathbb{R}^2} e^{i\mathbf{k}\cdot\mathbf{x}} K(\mathbf{x}, \varepsilon \mathbf{k}) \widehat{f}(\mathbf{k}) \frac{d\mathbf{k}}{(2\pi)^2}. \quad (11)$$

3 MATHEMATICAL TOOLS

This short section introduces the main mathematical tools used in the subsequent analysis for the derivation of the transport equation of Rayleigh waves from the two-scale boundary condition (10) defining their propagation at the surface of an homogeneous half-space.

3.1 Wigner transform and Wigner measure

The transport equation for the energy density of a high-frequency Rayleigh wave is derived starting from the Wigner distribution of its amplitude A_ε . It is a function of position $\mathbf{x}$ on the surface and wave vector $\mathbf{k}$ and it is scaled by the small parameter ε . The Wigner transform of $f, g \in \mathcal{S}'(\mathbb{R}^2)$ is defined by:

$$W_\varepsilon[f, g](\mathbf{x}, \mathbf{k}) = \int_{\mathbb{R}^2} e^{i\mathbf{k}\cdot\mathbf{y}} f\left(\mathbf{x} - \frac{\varepsilon\mathbf{y}}{2}\right) \overline{g\left(\mathbf{x} + \frac{\varepsilon\mathbf{y}}{2}\right)} \frac{d\mathbf{y}}{(2\pi)^2}$$

where $\bar{f}$ stands for the complex conjugate of f . Provided that the sequence A_ε lies in a bounded subset of $L^2(\mathbb{R}^2)$, the complex-valued sequence $W_\varepsilon[A_\varepsilon] := W_\varepsilon[A_\varepsilon, A_\varepsilon]$ has (up to an extracted subsequence) a real weak-* limit in $\mathcal{S}'(T^*\mathbb{R}^2)$ as $\varepsilon \rightarrow 0$ which is also a non-negative measure, the so-called Wigner measure W of A_ε (Gérard et al. 1997).

3.2 Some pseudo-differential calculus

In this section we give without proofs some useful results on pseudo-differential operators (see Bal 2005, Gérard et al. 1997 for some examples in the same context) which we shall need in the subsequent section for the derivation of the transport properties of the Wigner

measure W . First, we will have to compute the Wigner transform:

$$W_\varepsilon[K(\varepsilon D)f, g](\mathbf{x}, \mathbf{k}) = K\left(\mathbf{x} + \frac{\varepsilon D}{2}\right) W_\varepsilon[f, g](\mathbf{x}, \mathbf{k}) \quad (12)$$

for an homogeneous operator $K(D)$. This relation can be expanded in the form (Bal 2005, Gérard et al. 1997):

$$W_\varepsilon[K(\mathbf{x}, \varepsilon D)f, g](\mathbf{x}, \mathbf{k}) = K(\mathbf{x}, \mathbf{k}) W_\varepsilon[f, g] + \frac{\varepsilon}{2i} \{K, W_\varepsilon[f, g]\} - \frac{\varepsilon}{2i} \nabla_{\mathbf{x}} \cdot \nabla_{\mathbf{k}} K(\mathbf{x}, \mathbf{k}) W_\varepsilon[f, g] + O(\varepsilon^2) \quad (13)$$

for a non-homogeneous operator $K(\mathbf{x}, D)$, where $\{K, W\}$ stands for the usual Poisson bracket:

$$\{K, W\} = \nabla_{\mathbf{k}} K \cdot \nabla_{\mathbf{x}} W - \nabla_{\mathbf{x}} K \cdot \nabla_{\mathbf{k}} W.$$

To account for the highly oscillatory fluctuations of the random impedance we shall also use the relation (Bal 2005):

$$W_\varepsilon\left[\gamma\left(\mathbf{x}, \frac{\mathbf{x}}{\varepsilon}\right) K(\mathbf{x}, \varepsilon D)f, g\right](\mathbf{x}, \mathbf{k}) = \int_{\mathbb{R}^2} e^{i\frac{\mathbf{k}'\cdot\mathbf{y}}{\varepsilon}} \widehat{\gamma}(\mathbf{x}, \mathbf{k}') K\left(\mathbf{x}, \mathbf{k} - \frac{\mathbf{k}'}{2} + \frac{\varepsilon D}{2}\right) \times W_\varepsilon[f, g]\left(\mathbf{x}, \mathbf{k} - \frac{\mathbf{k}'}{2}\right) \frac{d\mathbf{k}'}{(2\pi)^2} + O(\varepsilon), \quad (14)$$

where $\widehat{\gamma}$ is the Fourier transform of γ with respect to the fast scale $\frac{\mathbf{x}}{\varepsilon}$.

4 TRANSPORT EQUATION FOR RAYLEIGH WAVES

This section now presents the main results of this research: the derivation of the radiative transport equation for the amplitude vector A_ε of the Rayleigh waves. The analysis starts from the Wigner-Liouville equation associated to the two-scale boundary condition (10) and is carried on using a multi-scale expansion of the Wigner transform of the sequence (A_ε) about its Wigner measure (the limit Wigner transform as ε goes to 0). Equating like powers of ε yields the dispersion properties of the Wigner measure, then the explicit expression of the lower-order corrector of the Wigner transform, and finally the radiative transfer equation itself. These developments basically follow the derivation in (Bal 2005, Bal et al. 2000, Bal and Ryzhik 2002, Papanicolaou and Ryzhik 1999, Powell and Vanneste 2005).

4.1 The Wigner-Liouville equation

Premultiplying Eq. (10) by A_ε^* , where f^* means complex conjugation and transposition if need be, then taking the Wigner transform and summing with the adjoint equation, one obtains the so-called Wigner-Liouville equation:

$$\begin{aligned} & W_\varepsilon[P(\varepsilon\mathbf{D})A_\varepsilon, A_\varepsilon] + W_\varepsilon[A_\varepsilon, P(\varepsilon\mathbf{D})A_\varepsilon] \\ &= \sqrt{\varepsilon}W_\varepsilon[Z(\mathbf{y})Q(\varepsilon\mathbf{D})A_\varepsilon, A_\varepsilon] \\ & \quad + \sqrt{\varepsilon}W_\varepsilon[A_\varepsilon, Z(\mathbf{y})Q(\varepsilon\mathbf{D})A_\varepsilon] \quad (15) \end{aligned}$$

where $\mathbf{y} := \frac{\mathbf{x}}{\varepsilon}$. The Wigner transform $W_\varepsilon[A_\varepsilon]$ of the sequence A_ε is now expanded in two scales $\mathbf{x}$ (the slow one) and $\mathbf{y}$ (the fast one) as (Papanicolaou & Ryzhik 1999):

$$\begin{aligned} W_\varepsilon[A_\varepsilon](\mathbf{x}, \mathbf{k}) &= W(\mathbf{x}, \mathbf{k}) \\ & \quad + \sqrt{\varepsilon}W_{\frac{1}{2}}(\mathbf{x}, \mathbf{y}, \mathbf{k}) + \varepsilon W_1(\mathbf{x}, \mathbf{y}, \mathbf{k}) + o(\varepsilon) \quad (16) \end{aligned}$$

and plugged into Eq. (15), which becomes¹ owing to Eq. (12):

$$\begin{aligned} & P\left(\mathbf{k} + \frac{\mathbf{D}_y}{2} + \frac{\varepsilon\mathbf{D}_x}{2}\right)W_\varepsilon[A_\varepsilon] + W_\varepsilon[A_\varepsilon]P^*\left(\mathbf{k} - \frac{\mathbf{D}_y}{2} - \frac{\varepsilon\mathbf{D}_x}{2}\right) \\ &= \sqrt{\varepsilon}(W_\varepsilon[Z(\mathbf{y})Q(\varepsilon\mathbf{D})A_\varepsilon, A_\varepsilon] \\ & \quad + W_\varepsilon[A_\varepsilon, Z(\mathbf{y})Q(\varepsilon\mathbf{D})A_\varepsilon]) \quad (17) \end{aligned}$$

since $\varepsilon\mathbf{D} = \varepsilon\mathbf{D}_x + \mathbf{D}_y$ whenever $\mathbf{y} = \frac{\mathbf{x}}{\varepsilon}$.

since $\varepsilon\mathbf{D} = \varepsilon\mathbf{D}_x + \mathbf{D}_y$ whenever $\mathbf{y} = \frac{\mathbf{x}}{\varepsilon}$.

4.2 Dispersion properties of the Wigner measure

The above Eq. (17) yields to the leading order $O(\varepsilon^0)$ the following dispersion equation:

$$P(\mathbf{k})W(\mathbf{x}, \mathbf{k}) + W(\mathbf{x}, \mathbf{k})P^*(\mathbf{k}) = 0.$$

Thus the Wigner measure $W(\mathbf{x}, \mathbf{k})$ reads:

$$W(\mathbf{x}, \mathbf{k}) = w(\mathbf{x}, \hat{\mathbf{k}})|\Omega'(\mathbf{k})|\delta(|\mathbf{k}| - k_R), \quad (18)$$

where w is a scalar positive measure on $S^*\mathbb{R}^2$ and Ω is defined by Eq. (9). Here $\Omega' := \frac{d\Omega}{dk}$ and $\Omega(\mathbf{k}) = 0$ is nothing but Rayleigh's dispersion equation (3).

¹ Notation $W(\mathbf{x}, \mathbf{k})P^*(\mathbf{k} - \varepsilon\mathbf{D})$ should be interpreted as the inverse Fourier transform of the function $\widehat{W}(\mathbf{k}', \mathbf{k})P(\mathbf{k} - \varepsilon\mathbf{k}')$, in accordance with the definition (11).

4.3 The $O(\sqrt{\varepsilon})$ corrector

To the order $O(\sqrt{\varepsilon})$ one has owing to Eq. (14):

$$\begin{aligned} & P\left(\mathbf{k} + \frac{\mathbf{D}_y}{2}\right)W_{\frac{1}{2}}(\mathbf{x}, \mathbf{y}, \mathbf{k}) + W_{\frac{1}{2}}(\mathbf{x}, \mathbf{y}, \mathbf{k})P^*\left(\mathbf{k} - \frac{\mathbf{D}_y}{2}\right) \\ &= \int_{\mathbb{R}^2} \frac{d\mathbf{k}'}{(2\pi)^2} e^{i\mathbf{k}' \cdot \mathbf{y}} \widehat{Z}(\mathbf{k}') \left[Q\left(\mathbf{k} - \frac{\mathbf{k}'}{2}\right) W\left(\mathbf{x}, \mathbf{k} - \frac{\mathbf{k}'}{2}\right) \right. \\ & \quad \left. + W\left(\mathbf{x}, \mathbf{k} + \frac{\mathbf{k}'}{2}\right) \overline{Q\left(\mathbf{k} + \frac{\mathbf{k}'}{2}\right)} \right] \end{aligned}$$

since W does not depend on the fast scale $\mathbf{y}$, or, if one considers the Fourier transform $\widehat{W}_{\frac{1}{2}}$ of $W_{\frac{1}{2}}$ with respect to that fast scale:

$$\begin{aligned} & P\left(\mathbf{k} + \frac{\mathbf{k}'}{2}\right)\widehat{W}_{\frac{1}{2}}(\mathbf{x}, \mathbf{k}', \mathbf{k}) + \widehat{W}_{\frac{1}{2}}(\mathbf{x}, \mathbf{k}', \mathbf{k})\overline{P\left(\mathbf{k} - \frac{\mathbf{k}'}{2}\right)} \\ &+ \theta\widehat{W}_{\frac{1}{2}}(\mathbf{x}, \mathbf{k}', \mathbf{k}) = \widehat{Z}(\mathbf{k}') \left[Q\left(\mathbf{k} - \frac{\mathbf{k}'}{2}\right) W\left(\mathbf{x}, \mathbf{k} - \frac{\mathbf{k}'}{2}\right) \right. \\ & \quad \left. + W\left(\mathbf{x}, \mathbf{k} + \frac{\mathbf{k}'}{2}\right) \overline{Q\left(\mathbf{k} + \frac{\mathbf{k}'}{2}\right)} \right]. \quad (19) \end{aligned}$$

Here $0 < \theta \ll 1$ is a small arbitrary parameter introduced for regularization that will be sent to 0 at the end of the derivation (Papanicolaou & Ryzhik 1999). Thus:

$$\widehat{W}_{\frac{1}{2}}(\mathbf{x}, \mathbf{k}', \mathbf{k}) = i\widehat{Z}(\mathbf{k}')\widehat{X}_\theta(\mathbf{x}, \mathbf{k}', \mathbf{k}) \quad (20)$$

with:

$$\begin{aligned} & \widehat{X}_\theta(\mathbf{x}, \mathbf{k}', \mathbf{k}) = \\ & \frac{Q(\mathbf{k} - \frac{\mathbf{k}'}{2})W(\mathbf{x}, \mathbf{k} - \frac{\mathbf{k}'}{2}) + \overline{Q(\mathbf{k} + \frac{\mathbf{k}'}{2})W(\mathbf{x}, \mathbf{k} + \frac{\mathbf{k}'}{2})}}{\Omega(\mathbf{k} - \frac{\mathbf{k}'}{2}) - \Omega(\mathbf{k} + \frac{\mathbf{k}'}{2}) + i\theta}. \end{aligned}$$

4.4 Derivation of the radiative transfer equation

To the first order $O(\varepsilon)$ one now has owing to Eq. (13) and again Eq. (14):

$$\begin{aligned} & \frac{1}{2i}(\{P, W\} + \{W, P^*\}) + P\left(\mathbf{k} + \frac{\mathbf{D}_y}{2}\right)W_1 \\ & \quad + W_1P^*\left(\mathbf{k} - \frac{\mathbf{D}_y}{2}\right) = I_\theta + I_\theta^* \quad (21) \end{aligned}$$

where:

$$\begin{aligned} I_\theta &= \int_{\mathbb{R}^2} \frac{d\mathbf{k}'}{(2\pi)^2} e^{i\mathbf{k}' \cdot \mathbf{y}} \widehat{Z}(\mathbf{k}') Q\left(\mathbf{k} - \frac{\mathbf{k}'}{2} + \frac{\mathbf{D}_y}{2}\right) \times \\ & \quad W_{\frac{1}{2}}\left(\mathbf{x}, \mathbf{y}, \mathbf{k} - \frac{\mathbf{k}'}{2}\right). \end{aligned}$$

We then consider the ensemble average of the above equation. First it is assumed that W_1 is orthogonal to W in order to justify the expansion (16), so that $\mathbb{E}\{PW_1 + W_1P^*\} = 0$. Next, using Eq. (20) one has:

$$\begin{aligned} I_\theta &= \int_{\mathbb{R}^4} \frac{d\mathbf{k}'d\mathbf{k}''}{(2\pi)^4} e^{i(\mathbf{k}'+\mathbf{k}'')\cdot\mathbf{y}} \hat{Z}(\mathbf{k}')Q\left(\mathbf{k}-\frac{\mathbf{k}'-\mathbf{k}''}{2}\right) \\ &\quad \times \hat{W}_{\frac{1}{2}}\left(\mathbf{x}, \mathbf{k}'', \mathbf{k}-\frac{\mathbf{k}'}{2}\right) \\ &= \int_{\mathbb{R}^4} \frac{i d\mathbf{k}'d\mathbf{k}''}{(2\pi)^4} e^{i(\mathbf{k}'+\mathbf{k}'')\cdot\mathbf{y}} \hat{Z}(\mathbf{k}')\hat{Z}(\mathbf{k}'')Q\left(\mathbf{k}-\frac{\mathbf{k}'-\mathbf{k}''}{2}\right) \\ &\quad \times \hat{X}_\theta\left(\mathbf{x}, \mathbf{k}'', \mathbf{k}-\frac{\mathbf{k}'}{2}\right). \end{aligned}$$

But $\mathbb{E}\{\hat{Z}(\mathbf{k}')\hat{Z}(\mathbf{k}'')\} = (2\pi)^2 \hat{R}(\mathbf{k}')\delta(\mathbf{k}'+\mathbf{k}'')$ by definition of the power spectral density $\hat{R}(\mathbf{k})$ of the mean-square stationary stochastic process $(Z(\mathbf{x}), \mathbf{x} \in \mathbb{R}^2)$, so that taking the ensemble average of I_θ yields:

$$\begin{aligned} \mathbb{E}\{I_\theta\} &= \int_{\mathbb{R}^2} \frac{i d\mathbf{k}'}{(2\pi)^2} \hat{R}(\mathbf{k}')Q(\mathbf{k}-\mathbf{k}')\hat{X}_\theta\left(\mathbf{x}, -\mathbf{k}', \mathbf{k}-\frac{\mathbf{k}'}{2}\right) \\ &= \int_{\mathbb{R}^2} \frac{i d\mathbf{k}'}{(2\pi)^2} \hat{R}(\mathbf{k}-\mathbf{k}')Q(\mathbf{k}')\hat{X}_\theta\left(\mathbf{x}, \mathbf{k}'-\mathbf{k}, \frac{\mathbf{k}+\mathbf{k}'}{2}\right) \end{aligned}$$

In the above calculation a crucial (mixing) assumption has been used: indeed it is expected that $\mathbb{E}\{ZZW\} \simeq \mathbb{E}\{ZZ\}\mathbb{E}\{W\}$ since both quantities Z and W vary on different scales; this assumption may be justified rigorously for the Schrödinger equation (Erdős & Yau 2000) or the scalar wave equation in a discrete setting (Lukkarinen & Spohn 2007). Note also that in the above equation and below $\mathbb{E}\{W\}$ is still denoted by W for convenience. Then we obtain:

$$\begin{aligned} \mathbb{E}\{I_\theta + I_\theta^*\} &= \\ &\int_{\mathbb{R}^2} \frac{i d\mathbf{k}'}{(2\pi)^2} \hat{R}(\mathbf{k}-\mathbf{k}')|Q(\mathbf{k}')|^2 (W(\mathbf{x}, \mathbf{k}') - W(\mathbf{x}, \mathbf{k})) \\ &\quad \times \left(\frac{1}{\Omega(\mathbf{k}) - \Omega(\mathbf{k}') + i\theta} - \frac{1}{\Omega(\mathbf{k}) - \Omega(\mathbf{k}') - i\theta} \right) \\ &\quad - 2W(\mathbf{x}, \mathbf{k}) \int_{\mathbb{R}^2} \frac{d\mathbf{k}'}{(2\pi)^2} \hat{R}(\mathbf{k}-\mathbf{k}')\hat{Y}_\theta(\mathbf{k}', \mathbf{k}), \end{aligned}$$

where:

$$\hat{Y}_\theta(\mathbf{k}', \mathbf{k}) = \Im \left\{ \frac{Q(\mathbf{k})Q(\mathbf{k}') + |Q(\mathbf{k}')|^2}{\Omega(\mathbf{k}) - \Omega(\mathbf{k}') + i\theta} \right\}.$$

The above integrals are split in four parts, for $|\mathbf{k}'| \geq k_R$, $k_s \leq |\mathbf{k}'| < k_R$, $k_p \leq |\mathbf{k}'| < k_s$, and $|\mathbf{k}'| < k_p$. As $\mathbf{k}$ is such that $|\mathbf{k}| = k_R$ because the Wigner measure $W(\mathbf{x}, \mathbf{k})$ has

support on that circle, $\Omega(\mathbf{k})$ is real and $Q(\mathbf{k})$ is purely imaginary. $Q(\mathbf{k}')$ is purely imaginary whenever $|\mathbf{k}'| > k_p$, and $\Omega(\mathbf{k}')$ is real whenever $|\mathbf{k}'| \geq k_s$ or $|\mathbf{k}'| \leq k_p$; thus the second integral reduces to:

$$\begin{aligned} &- 2 \int_{|\mathbf{k}'| \geq k_R} \frac{d\mathbf{k}'}{(2\pi)^2} \hat{R}(\mathbf{k}-\mathbf{k}')\hat{Y}_\theta(\mathbf{k}', \mathbf{k}) \\ &- 2 \int_{|\mathbf{k}'| < k_R} \frac{d\mathbf{k}'}{(2\pi)^2} \hat{R}(\mathbf{k}-\mathbf{k}')\hat{Y}_\theta(\mathbf{k}', \mathbf{k}) \\ &\rightarrow \int_{|\mathbf{k}'| \geq k_R} \frac{d\mathbf{k}'}{2\pi} \hat{R}(\mathbf{k}-\mathbf{k}') (Q(\mathbf{k})Q(\mathbf{k}') + |Q(\mathbf{k}')|^2) \\ &\quad \times \delta(\Omega(\mathbf{k}) - \Omega(\mathbf{k}')) \\ &- 2 \int_{|\mathbf{k}'| \leq k_s} \frac{d\mathbf{k}'}{(2\pi)^2} \hat{R}(\mathbf{k}-\mathbf{k}')\hat{Y}_0(\mathbf{k}', \mathbf{k}) \end{aligned}$$

in the limit $\theta \rightarrow 0$. Also $W(\mathbf{x}, \mathbf{k}') = 0$ whenever $|\mathbf{k}'| < k_R$, so that the first integral reduces to:

$$\begin{aligned} &\int_{|\mathbf{k}'| \geq k_R} \frac{i d\mathbf{k}'}{(2\pi)^2} \hat{R}(\mathbf{k}-\mathbf{k}')|Q(\mathbf{k}')|^2 (W(\mathbf{x}, \mathbf{k}') - W(\mathbf{x}, \mathbf{k})) \\ &\quad \times \left(\frac{1}{\Omega(\mathbf{k}) - \Omega(\mathbf{k}') + i\theta} - \frac{1}{\Omega(\mathbf{k}) - \Omega(\mathbf{k}') - i\theta} \right) \\ &+ 2W(\mathbf{x}, \mathbf{k}) \int_{|\mathbf{k}'| < k_R} \frac{d\mathbf{k}'}{(2\pi)^2} \Im \left\{ \frac{\hat{R}(\mathbf{k}-\mathbf{k}')|Q(\mathbf{k}')|^2}{\Omega(\mathbf{k}) - \Omega(\mathbf{k}') + i\theta} \right\} \\ &\rightarrow \int_{|\mathbf{k}'| \geq k_R} \frac{d\mathbf{k}'}{2\pi} \hat{R}(\mathbf{k}-\mathbf{k}')|Q(\mathbf{k}')|^2 \delta(\Omega(\mathbf{k}) - \Omega(\mathbf{k}')) \\ &\quad \times (W(\mathbf{x}, \mathbf{k}') - W(\mathbf{x}, \mathbf{k})) \end{aligned}$$

$$+ 2W(\mathbf{x}, \mathbf{k}) \int_{|\mathbf{k}'| < k_s} \frac{d\mathbf{k}'}{(2\pi)^2} \Im \left\{ \frac{\hat{R}(\mathbf{k}-\mathbf{k}')|Q(\mathbf{k}')|^2}{\Omega(\mathbf{k}) - \Omega(\mathbf{k}')} \right\}$$

as $\theta \rightarrow 0$. Since the left-hand-side in Eq. (21) is:

$$\begin{aligned} \frac{1}{2i} \mathbb{E}\{\{P, W\} + \{W, P^*\}\} &= \nabla_{\mathbf{k}} \Omega \cdot \nabla_{\mathbf{x}} W \\ &= \Omega'(\mathbf{k}) \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} W, \end{aligned}$$

and $Q(\mathbf{k}) = Q(\mathbf{k}')$ if $|\mathbf{k}| = |\mathbf{k}'|$, one finally gets the transport equation:

$$\begin{aligned} &\left(\Omega'(\mathbf{k}) \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} + \eta(\mathbf{k}) \right) W(\mathbf{x}, \mathbf{k}) = \\ &\int_{|\mathbf{k}'| = k_R} \frac{d\mathbf{k}'}{2\pi} \hat{R}(\mathbf{k}-\mathbf{k}') \overline{Q(\mathbf{k}')} Q(\mathbf{k}) \delta(\Omega(\mathbf{k}') - \Omega(\mathbf{k})) \\ &\quad \times (W(\mathbf{x}, \mathbf{k}') - W(\mathbf{x}, \mathbf{k})) \quad (22) \end{aligned}$$

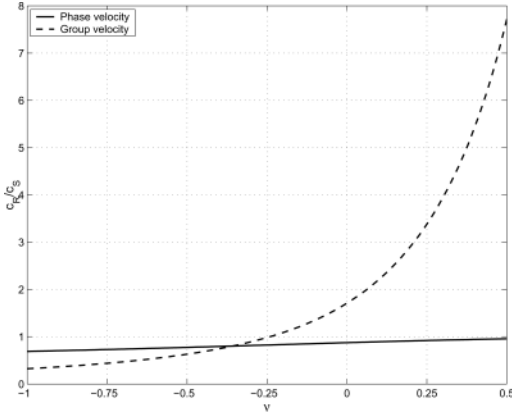


Figure 1. Normalized phase () and transport velocities of Rayleigh waves as functions of Poisson's coefficient ν .

for the Wigner measure $\mathcal{W}$ of Rayleigh waves amplitude, with:

$$\eta(\mathbf{k}) = \int_{|\mathbf{k}'| \leq k_s} \frac{d\mathbf{k}'}{2\pi^2} \hat{R}(\mathbf{k} - \mathbf{k}') \Im \left\{ \frac{Q(\mathbf{k})Q(\mathbf{k}')}{\Omega(\mathbf{k}) - \Omega(\mathbf{k}')} \right\}.$$

The right-hand-side of Eq. (22) is the scattering operator of Rayleigh surface waves into themselves with the same wave numbers $|\mathbf{k}| = |\mathbf{k}'| = k_R$ but different directions. The second term on the left-hand-side accounts for leakage of Rayleigh waves into bulk waves which propagate away from the surface in the lower half-space. It is a purely lossy term because the half-space is homogeneous and thus bulk waves do not return to the interface. Indeed, one can easily check from (9) and (8) that:

$$\Im \left\{ \frac{Q(\mathbf{k})Q(\mathbf{k}')}{\Omega(\mathbf{k}) - \Omega(\mathbf{k}')} \right\} > 0$$

whenever $|\mathbf{k}'| \leq k_s$. As $\Omega(\mathbf{k}) = 0$ on the support of $\mathcal{W}(\mathbf{x}, \mathbf{k})$, the loss factor is:

$$\eta(\mathbf{k}) = - \int_{|\mathbf{k}'| \leq k_s} \frac{d\mathbf{k}'}{2\pi^2} \hat{R}(\mathbf{k} - \mathbf{k}') \Im \left\{ \frac{Q(\mathbf{k})Q(\mathbf{k}')}{\Omega(\mathbf{k}')} \right\}, \quad (23)$$

where $|\mathbf{k}| = k_R$. At last, the first term on the left-hand-side is the streaming flow of energy along straight rays on the surface at the speed $\Omega'(\mathbf{k})$ for $|\mathbf{k}| = k_R$. This normalized transport velocity is plotted on Fig. (1) as a function of the half-space Poisson's ratio $-1 \leq \nu \leq \frac{1}{2}$. Rayleigh waves phase speed c_R as given by Eq. (3) is also displayed (normalized by c_s).

4.5 Influence of the impedance power spectrum

The strength of scattering and leakage of Rayleigh surface waves may be compared as in Bal et al.

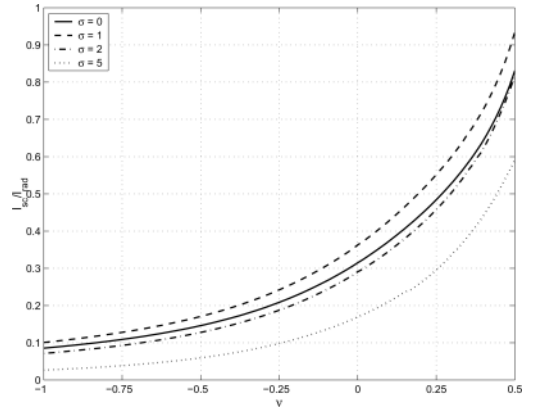


Figure 2. The ratio ℓ/ℓ_{rad} as function of Poisson's coefficient ν for $k_0 = 0$.

(2000) introducing the scattering and radiation lengths defined by:

$$\ell^{-1} = \int_{|\mathbf{k}'| = k_R} \frac{d\mathbf{k}'}{2\pi} \hat{R}(\mathbf{k} - \mathbf{k}') \overline{Q(\mathbf{k})} Q(\mathbf{k}') \delta(\Omega(\mathbf{k}')) \quad (24)$$

and $\ell_{\text{rad}}^{-1} := \eta$, respectively. For a statistically isotropic random impedance $(Z(\mathbf{x}), \mathbf{x} \in \mathbb{R}^2)$, such that its power spectral density $\hat{R}(\mathbf{k})$ depends on $|\mathbf{k}|$ solely, these parameters are:

$$(k_R \ell)^{-1} = \frac{|Q(\mathbf{k})|^2}{\Omega'(\mathbf{k})} \int_0^{2\pi} \frac{d\phi}{2\pi} \hat{R}(k_R \sqrt{2 - 2 \cos \phi})$$

where $|\mathbf{k}| = k_R$, and:

$$(k_R \ell_{\text{rad}})^{-1} = \frac{k_s^4 \sqrt{k_R^2 - k_p^2}}{2\pi^2 k_R^2 (2k_R^2 - k_s^2)} \times \left[\int_0^{k_p} \int_0^{2\pi} \frac{k \sqrt{k_p^2 - k^2} \hat{S}(k, \phi) d\phi dk}{(2k^2 - k_s^2)^2 + 4k^2 \sqrt{(k^2 - k_p^2)(k^2 - k_s^2)}} + 4 \int_{k_p}^{k_s} \int_0^{2\pi} \frac{k^3 (k^2 - k_p^2) \sqrt{k_s^2 - k^2} \hat{S}(k, \phi) d\phi dk}{(2k^2 - k_s^2)^4 - 16k^4 (k^2 - k_p^2)(k^2 - k_s^2)} \right],$$

where $\hat{S}(k, \phi) = \hat{R}(\sqrt{k^2 + k_R^2 - 2kk_R \cos \phi})$. The ratio ℓ/ℓ_{rad} is plotted on Fig. (2) through Fig. (4) as a function of ν for a Gaussian model of correlation:

$$\hat{R}(\mathbf{k}) = e^{-\sigma(|\mathbf{k}| - k_0)^2}, \quad (25)$$

where $\sqrt{\sigma}$ is the correlation length of the random impedance and k_0 is its central frequency. The case $\sigma = 0$ corresponds to a delta-correlated impedance. One observes that radiation dominates over scattering for positive values of Poisson's coefficient. An efficient means to increase scattering over radiation is to

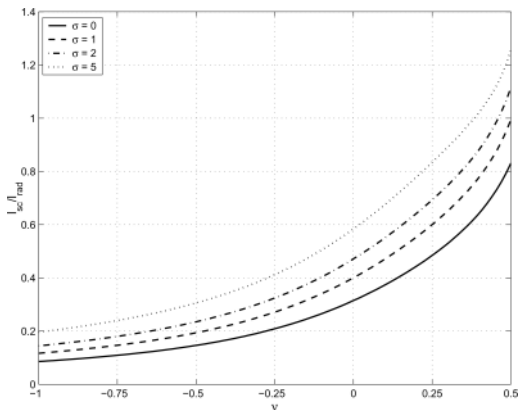


Figure 3. The ratio ℓ/ℓ_{rad} as function of Poisson's coefficient ν for $k_0 = k_R$.

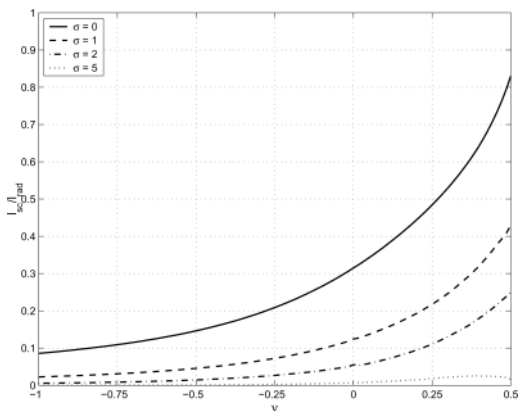


Figure 4. The ratio ℓ/ℓ_{rad} as function of Poisson's coefficient ν for $k_0 = 3k_R$.

increase the central frequency of the spectrum (25). Similar conclusions may be reached for an exponential model of correlation, for example, corresponding to a mean-square non-differentiable random impedance process.

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